

Cpp_programs\Robocon_programs\Assignment_3_day_4.cpp

```
1  /*
2  3. Problem Statement:
3
4  You are designing a simple control system for a robotic vehicle using an Arduino. The robot has two
  switches that control its movement. Three LEDs – Green, Yellow, and Red – are used to indicate what
  the robot is currently doing.
5
6  Your task is to connect the two switches and three LEDs to the Arduino and write a program that
  reads the state of both switches and controls the LEDs according to the following conditions:
7
8  • When both switches are ON:
9    - Green LED should be ON.
10   - Yellow LED should be OFF.
11   - Red LED should be OFF.
12   - The robot is moving forward.
13
14  • When the first switch is OFF and the second switch is ON:
15   - Green LED should be OFF.
16   - Yellow LED should be OFF.
17   - Red LED should be ON.
18   - The robot is moving backward.
19
20  • When the second switch is OFF, regardless of the state of the first switch:
21   - Green LED should be OFF.
22   - Yellow LED should be ON.
23   - Red LED should be OFF.
24   - The robot is stopped.
25
26  Implement a tinkercad circuit for the above problem statement
27
28  code by Rayan Raphy
29  */
30  int led_1 = 8;
31  int led_2 = 9;
32  int led_3 = 10;
33
34  int sw_1 = 2;
35  int sw_2 = 3;
36
37  void setup()
38  {
39    pinMode(led_1, OUTPUT);
40    pinMode(led_2, OUTPUT);
41    pinMode(led_3, OUTPUT);
42
43    pinMode(sw_1, INPUT);
44    pinMode(sw_2, INPUT);
45    Serial.begin(9600);
```

```
46  }
47
48  void loop()
49  {
50      bool sw_1_state = digitalRead(sw_1);
51      bool sw_2_state = digitalRead(sw_2);
52      digitalWrite(led_1,(sw_1_state && sw_2_state));//The robot is moving forward.
53      digitalWrite(led_2,!sw_2_state);//The robot is stopped.
54      digitalWrite(led_3,!sw_1_state && sw_2_state);//The robot is moving backward.
55  }
```